

Wahyu Ikbal Maulana

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SUMMARY

Fifth-semester Data Science undergraduate student with over 2 years of experience in Artificial Intelligence, Data analysis, and Software Development. Actively contributed to 5+ faculty research projects since the first semester and engaged in 7+ teaching activities in both professional and academic settings. Passionate about creating impact through teaching, creating educational content, and writing in the field of technology, while focusing on developing leadership skills through various opportunities.

EXPERIENCE

Lead AI Engineer

PT PAL Indonesia x Politeknik Elektronika Negeri Surabaya

September 2025 – Present

- Led the end-to-end design and implementation of an AI meeting assistant architecture in an industry collaboration PT PAL with PENS campus, using LangGraph and PGVector, integrated with React Router, and supported by a RabbitMQ-powered data pipeline, fully built on open-source technologies to meet client security and transparency requirements.
- Developed backend services for RAG-based meeting transcripts and summaries with open-source models served via VLLM and Langgraph Typescript Backend services, currently evaluating LLM output quality to achieve above 80% ROUGE score on outputs using LLM-as-a-judge.
- Managed workflows for a 5-person team and tailored AI applications to align with stakeholder requirements, balancing leadership responsibilities with hands-on engineering tasks.

AI Engineer

Perfect10 AI

Aug 2025 – Oct 2025

- Developed and deployed an AI documentation generator using LangGraph-based AI agents with parallelized workflows, integrated into the frontend and supported by FastAPI backend services.
- Conducted observability and experimentation on existing AI agents using LangSmith and LLM-as-a-judge to detect hallucinations and prompt system weaknesses, analyzing results in structured spreadsheets to break down scenarios and failure patterns, and collaborating with senior engineers for remediation.

AI Trainer

Dentech Corporation id

July 2025 – Present

- Organized and delivered over 4 training sessions in Artificial Intelligence and data science, covering topics such as Python for Machine Learning, AI Agents, LLMs, LangChain, and LangGraph, for audiences of practitioners and university students attended by dozens and reaching hundreds of viewers.
- Designed and developed structured learning materials, hands-on code resources, project-based case studies, and facilitated interactive Q&A sessions to support effective participant learning.

AI Engineer

80&Company

April 2025 – Sept 2025

- Developed a full-stack AI application using Next.js and Supabase, integrating the Dify API to automate the screening and classification of freelance engineers based on job relevance and quality, improving the efficiency of the company's recruitment process.
- Built a Selenium-based data scraping system integrated with a fine-tuned local LLM to analyze internal data across Japanese prefectures, supporting real-time insights for client business needs.

Freelance AI/ML Engineer

Individual/Self employed

February 2025 – Present

- Designed and implemented an automated MySQL database reporting system using Deno for a coffee export company, enabling easier and faster access to operational and financial information.
- Developed a RAG-based chatbot for an international company's internal knowledge base using LlamaIndex, implementing document chunking, embedding indexing, and vector retrieval to support cross-language communication and accurate understanding of internal knowledge.
- Assisted a Master's student from ITS in a brain tumor research project by providing guidance on deep learning architecture design and model structuring, delivered through remote collaboration using TeamViewer.

- Led knowledge transfer initiatives by training IT leaders to develop Langchain and Langgraph-based HR assistants, empowering clients to build in-house AI capabilities and improve HR processes.
- Improved the evaluation accuracy of a traditional RAG chat system by transforming it into an agentic RAG architecture using LangChain and the OpenAI API, increasing performance from 50% to 80%+ through optimal chunking strategy selection, retrieval strategy optimization, and prompt refinement.

Data Analyst Staff

PT. Media Karya Prestige Internasional (Kanotaria)

July 2024 - December 2024

- Processed and analyzed company data to create comprehensive company reports, leading to improved decision-making.
- Created a visualization dashboard that contributed to an improvement in cross-team efficiency and overall productivity.

Data & AI Researcher

Politeknik Elektronika Negeri Surabaya

December 2023 - Present

- Contributed to faculty-led AI research and community service projects with lecturers, developing dashboards with deep learning models and working on chatbot systems, text-to-SQL fine-tuning, and LLM-based use cases, and conducting cybersecurity-focused AI research, resulting in academic outputs including a book and conference posters.
- Conceptualized and organized AI webinars for 50+ participants to educate audiences on LLMs and RAG workflows, which expanded professional networking opportunities and led to multiple project collaborations.
- Serving as a laboratory assistant, managing and maintaining the operations of the artificial intelligence (AI) and data science laboratory, assisting several students with their final projects, and ensuring smooth access to computing resources and research infrastructure.

PROJECT

Bisawebsite | [Github](#), [Link](#)

- Developed an AI-powered website generator using a full TypeScript stack (Next.js and OpenAI). The platform enables anyone to build and deploy websites for free, is fully open-source, earned significant GitHub stars, and has attracted thousands of weekly visits with dozens of user sign-ups.

Open source Fastapi AI production | [Github](#)

- Developed an open-source project demonstrating how to run AI in production using popular stacks such as FastAPI, LangChain, and LangSmith, complete with containerization and Docker-based deployment, and received nearly 100 GitHub stars.

Ngumpulin | [Github](#), [Link](#)

- Developed an AI application using Next.js and PostgreSQL, integrating Machine Learning algorithms and Generative AI from the Llama model with a FastAPI backend to assist teachers in grading assignments.
- Streamlined task submissions for students while providing automated grading suggestions through an AI agent capable of processing and analyzing assignment content, delivering intelligent feedback based on predefined rubrics and contextual understanding.
- Implemented MLOps for automatic clustering of student assignments using Docker & MLflow, further enhancing the feedback mechanism and overall system performance into the database.

Malware Detection using Ensemble Method | [Github](#)

- Developed a malware detection system using several traditional machine learning techniques, and combined and evaluated several ensemble methods to improve classification accuracy and effectively identify malicious software up to 80%.
- Collaborated as part of a team to design the solution and address the problem, with the work documented in a research paper.

SKILLS

Programming & Tools: Python, R, Next JS, PostgreSQL, MySQL, Tailwind CSS, Typescript, Git, GitHub, Drizzle ORM, Prisma, Docker, Trello, Notion, Supabase, Tensorflow, Pytorch, Django, Fastapi, Flask, Deno, Expo React Native

Data science: Data science pipeline (cleansing, feature engineering, visualization, data modeling), Statistics, NLP, CV, Time series, Tableau, Power BI, Grafana, Mage ai, Apache Airflow, Apache Spark, Pytorch, Pentaho

Soft skills: Communication, Problem Solving, Critical Thinking, Adaptability, Project Management, Time Management, Flexibility and Agility, Leadership, Mentoring & Coaching.

EDUCATION

Undergraduate in Applied Data Science
Politeknik Elektronika Negeri Surabaya, GPA 3.68/4.00

Surabaya
Aug 2023– Present

ORGANIZATIONAL INVOLVEMENT & VOLUNTEER WORK

Founder

AI Research Community

Surabaya
February 2025 – Present

- Founded and led an AI research community on WhatsApp with over 500 members, sharing the latest insights on AI development, open-source tools, and AI services to support continuous learning and knowledge exchange among aspiring AI & ML engineers.

Data Science Content Writer

Data Wizard

Surabaya
October 2024 – December 2024

- Created educational content and engaging data science memes on Instagram to simplify complex concepts and inspire interest in data science for a broad audience.
- Collaborated as part of a team to design the solution and address the problem, with the work documented in a research paper.

UI UX Designer

E-CODE HIMIT

Surabaya
January 2024 – Feb 2025

- Conduct UI/UX research to identify user needs, pain points, and opportunities for improvement in the Web HIMA IT Apps and Web HIMIT Champion platforms.
- Redesign and enhance the user interface (UI) of the Web HIMA IT Apps to optimize usability, accessibility, and visual appeal for seamless activity management.
- Design and refine the UI of the Web HIMIT Champion platform, ensuring an intuitive and engaging user experience that aligns with user expectations and project goals.

Blog Writer

Himpunan Mahasiswa Teknik Informatika PENS

Surabaya
September 2024 – Present

- Authored and published 5+ educational blogs and articles on machine learning, and artificial intelligence on Medium, reaching a wide audience and enhancing technical knowledge among peers.

EEPIS Data Enthusiast Community

Politeknik Elektronika Negeri Surabaya

Surabaya
February 2025 – Present

- Mentored students in the community by fostering bonding through interactive sessions, answering questions, and providing tailored advice on data science topics, course materials, and academic strategies at Politeknik Elektronika Negeri Surabaya.

CERTIFICATION & COURSE

- Certification AI Engineer for Data Science Associate, Datacamp, 2025
- Data Science Associate, Datacamp, 2025
- AI Engineer Bootcamp, 365 Careers, 2024
- The Data Literacy Course: Learn How to Work With Data, 365 Careers, 2024
- Python and R for Scientific Research Data Analysis, Udemy, 2024
- IT Literacy Training, UICI Research & Community Service Institute, 2022
- Team Collaboration in the Technology Industry, Future skills Indonesia, 2023
- Python bootcamp, Udemy, 2021